

Book of abstracts

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Mitigating the impacts of wind energy in forests for bats: Potential and limitations

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The growing use of wind energy in Europe is leading to increased deployment of wind turbines in structurally rich habitats such as forests, which could affect legally protected bats. On the one hand, forest-dwelling bats lose habitats when forests are cleared for the construction and operation of wind turbines. On the other hand, clearings around turbines offer new habitats for species at high risk of collision, such as aerial insectivorous bats. Therefore, fatality rates at wind turbines in forests are expected to be high. Potentially, bat fatalities in forests can be mitigated by curtailment, i.e. temporary suspension of turbine operation during periods of high bat activity. However, its effectiveness has not been thoroughly investigated throughout Europe. Recently, we have observed forest bats avoiding wind turbines over up to 500 m distances. This avoidance is associated with turbine operation at relatively high wind speeds, suggesting that factors associated with turbine operation, such as noise emissions, are causing the avoidance in bats. In summary, wind turbines in forests cause several problems for bats, highlighting the need for wind turbines to be sited outside forests or in structurally poor forests. Curtailment practice seems key for naturally inclusive wind energy production in forests.

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Wildlife behavioural plasticity contributes to fine-scale spatio-temporal avoidance of humans

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Anthropogenic disturbance elicits adaptive responses in wildlife aimed at risk-avoidance, ultimately imposing constraints on their spatial and temporal niche. While long-term adaptive responses to humans have been well documented, short-term responses to fine-scale variations in human presence have been overlooked.

Here we estimate short-term deviations from local habitat use and diel activity in response to rapid variation in human disturbance. We employed an year-long intense camera-trapping within a small private reserve, and recorded spatio-temporal information on all sources of anthropogenic disturbance in the area. We then explored fine-scale avoidance strategies in 7 mammal species, while controlling for their average habitat use and diel activity.

We found species to avoid humans on either the spatial or temporal axis. Foxes, pine and beech marten, roe deers, and porcupines favored temporal avoidance, whereas badgers favored spatial avoidance. Notably, foxes displayed spatial attraction, likely due to anthropogenic trophic resources. Wild boars, however, did not display human-avoidance behaviors.

Our study highlights the role of behavioural plasticity in adjusting daily habitat use and activity patterns to varying levels of human presence over space and time, emphasizing the species-specific nature of responses to human disturbances, contributing to our understanding of behavioral strategies that allow wildlife-human coexistence.

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Comparing nature conservation policies using a Bayesian decision network: modeling sequential effects on adoption, farming practices, and biodiversity

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Identifying and implementing policy measures to reduce on-farm biodiversity loss requires a better understanding of complex system interactions. Studies on policy challenges to enhance on-farm biodiversity can be divided into those focusing on biodiversity-enhancing practices on farms versus those focusing on the success of policies in terms of farmers adoption rates. In this study, we combine both parts within an ad-hoc decision analytic method to predict which policy-scenario is most likely to improve biodiversity on smallholder farms across the European Union. We apply a Bayesian decision network (BDN) to assess on-farm biodiversity outcomes under five

policy scenarios including a) regulatory, b) market-based, c) incentive-based, d) voluntary information-driven, and e) policy mix. We collect and combine information from existing literature along with expert elicitation from a workshop held with decision makers and other stakeholders. We found the highest probability of positive biodiversity outcomes (48%) under a regulatory policy but the result was strongly dependent on the expected rate of adoption. We showcase a novel approach combining knowledge from different knowledge strains and sources that accounts for layers of uncertainty while informing biodiversity policy decisions.

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ID: 254

A Systematic Map of the evidence for mesopredator release in terrestrial mammalian systems.

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Mesopredator release is the name given to the ecological cascade process that occurs when the local decline in population of a dominant carnivore causes the increase in population of a subordinate mesopredator as a result of being released from top-down pressure. This can have widespread impact on the local ecosystems, however it has been shown that the process may not be as widespread as commonly thought. Here we seek to look for global trends in the literature and compare the occurrence of mesopredator release across

different systems with different carnivore community species composition. We reviewed peer-reviewed publications on the Web of Science Core Collection, BIOSIS Citation Index, Zoological Record, Scopus, and Google Scholar databases to collate the literature on mesopredator release from terrestrial systems around the world. This resulted in 5224 unique results after de-duplicating, of these 509 were referring to terrestrial mammals and monitored the abundance or distribution of two or more carnivores.

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Protected areas in Tunisia between biodiversity conservation and territorial management

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One of the characteristics of forests in Tunisia is the fact that they are inhabited. The local population, despite the local way of life acquired over the centuries, voluntarily or involuntarily exerts anthropozoic pressure on forest and natural resources which are already fragile and are facing several threats, namely fires and the effects of climate change. In protected areas, the problem is amplified in view of the objectives of creating these areas, which all converge towards the conservation of fragile ecosystems and the protection of endangered species of fauna and/or flora. In consultation with local populations, scientists and experts have developed management methods that both meet conservation objectives and take into consideration the needs of populations living there. In the best contexts, local people, aware of the importance of resources and ecosystems, become in turn protectors and managers of resources and protected areas. Facing current declining environmental, social and governance situation, a management model must be put in place in Tunisia which draws inspiration from its history and from its regional and international experience, and that congregates both the growing needs of inhabitants and users of forest areas, and the objectives of conservation and enhancement of ecosystems and species.

ID: 263

Comparative landscape genetics of two partly sympatric hare species

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In France, the mountain hare (*Lepus timidus*) distribution is restricted to alpine environments. Confined to high altitudes, mountain hare populations are biogeographically isolated and recent evidences indicate a decline in their spatial range. This decrease is most probably due to global warming and human activities, but an additional threat might come from the competition with the European hare (*L. europaeus*). The range of the two species overlap over an area from 1000 to 2000 m asl, and the larger European hare is supposed to outcompete and displace the smaller mountain hare. Maintenance of mountain hare genetic variation is of critical importance since low levels will limit its ability to respond to these different threats. As this maintenance is partly depending on gene flow within and among populations, we aimed to characterize the environmental determinants of the gene flow of both species and to assess whether the inter-specific competition impact it or not. We used a machine-learning landscape genetics approach on > 1000 individuals of each species covering most of the French Alps and genotyped at 12 microsatellites loci to identify the main variables affecting connectivity. Results will be used to inform management strategies for the conservation of the mountain hare.